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Adding vibration to high-intensity intervals increase time at high oxygen uptake in well-trained cyclists

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The importance of accumulated time ≥90% of maximal oxygen consumption (VO_{2max}) to improve performance in well-trained endurance athletes is well established. The present study compared the acute effects of adding vibrations (VIB; 40 Hz) with the work intervals during a high-intensity cycling session (HIT) with a traditional HIT session without vibration (TRAD) on time ≥90% of VO_{2max}, time ≥90% of peak heart rate (HR_{peak}), electromyography (EMG) activity, and mean power in well-trained cyclists (n = 10, VO_{2max}=78.6 ± 7.4 mL/min/kg). The order of VIB and TRAD was randomized and consisted of 6 × 5-minutes work intervals performed with the highest possible mean power across the work intervals (2.5-minutes standardized relief periods). VIB was superior to TRAD on time ≥90% of VO_{2max}, (10.99 ± 7.00 vs 6.95 ± 5.28 minutes, respectively), time ≥90% of HR_{peak} (24.61 ± 2.38 vs 19.97 ± 4.12 minutes, respectively), and averaged EMG activity in m. Vastus Lateralis during the work intervals (all $P < 0.05$). The EMG/power output ratio across all work intervals was higher in VIB than TRAD ($P < 0.05$). Mean values across work intervals showed no difference between VIB and TRAD in mean power, rate of perceived exertion, or blood lactate concentration. Thus, the present study indicated that adding vibration to the work intervals during a HIT session can acutely increase the physiological responses of the cardiovascular system and increase time ≥90% VO_{2max} and should therefore be considered in order to optimize the exercise stimulus of well-trained cyclists.

KEYWORDS

cycling performance, cycling training, endurance, intense cycling exercise

1 | INTRODUCTION

There seems to be a consensus in the research literature that well-trained endurance athletes should perform a certain amount of training at intensities of 90%-100% of maximal oxygen uptake (VO_{2max}).[1-5] Moreover, it has been suggested that the training time ≥90% VO_{2max} could serve as a good criteria to judge the effectiveness of the stimulus to improve aerobic fitness.[2,6] However, continuous work at such high intensities cannot be sustained for a long time and thus limits total training time at this intensity during a single training session. Therefore, there is a quest for developing high-intensity aerobic interval training (HIT) sessions that optimizes time ≥90% VO_{2max}.[5] Accordingly, it has recently been observed that the HIT session that elicits the highest time at high VO_2 induces the largest training adaptation.[7]

A potential alternative to increase the stimulus during a HIT session is to add vibration to the exercise. Vibration is here defined as a mechanical stimulus that can affect the musculo-skeletal system by increasing the muscle activity.[8] The primary afferent endings of the muscle spindles are sensitive to vibration.[9] Therefore, has it been suggested

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that vibration increases the activation of stretch-sensitive muscle spindles ensuring increased Ia afferent signals and thus a larger excitatory inflow to the motoneurone pool and possibly increased muscle activation.[8,10,11] It is possible to add vibration while performing cycling exercise by mounting a cycle to a vibration plate [10] with concomitant increase in muscle activation.[12] Interestingly, only two studies have investigated the acute effects of adding vibrations while cycling at the high workloads used for HIT sessions, and they both observed increased VO_2 compared to cycling without vibration.[10,13] However, the participants in those two studies were only recreationally to moderate trained (mean VO_{2max} <57 mL/min/kg) and not specifically trained for cycling. Furthermore, both studies performed only one incremental test to exhaustion making it difficult to tell whether adding vibrations to a typical HIT session would increase time ≥90% VO_{2max} and how it would affect the ones who are pursuing the marginal gains; the well-trained cyclists. Actually, for almost 30 years ago, it was observed that the time to exhaustion at a constant workload was reduced from 60 to 47 minutes when vibration was added to submaximal cycling.[14] Adding vibrations to a HIT session might therefore accelerate the development of fatigue and thereby reduce the power output, the mechanical training stimulus and maybe also reduce time ≥90% VO_{2max} or on the other hand, it could increase the muscle activity and VO_2 and thus time ≥90% VO_{2max}.

Therefore, the primary purpose of the present study was to compare the acute effect of an all-out traditional HIT session without vibration (TRAD) with the same all-out HIT session with added vibration during the work intervals (VIB) on time ≥90% VO_{2max}, muscle activation and power output in well-trained cyclists.

2 | METHODS

2.1 | Subjects

Twelve male cyclists volunteered for the study, which was performed according to the ethical standards established by the Helsinki Declaration of 1975 and was approved by The local ethical committee of the Department of Sports Science, Lillehammer University College. All cyclists signed an informed consent form prior to participation. Two of the cyclists did not complete the study due to illness during the intervention period and their data are excluded (n = 10, age 24 ± 7 years, height 180 ± 5 cm, body mass 73.2 ± 6.2 kg, VO_{2max} 78.6 ± 7.4 mL/min/kg, maximal aerobic power 471 ± 29 W, maximal heart rate (HR_{max}) 195 ± 5 beats/min). The self-reported amount of training hours during the year preceding the intervention was 653 ± 83 hour. The endurance training was categorized into three heart rate (HR) zones: (1) 60%-82%, (2) 83%-87%, and (3) 88%-100% of maximal heart rate. The mean weekly intensity distribution into training zone 1 (low intensity), zone 2 (moderate intensity), and zone 3 (high intensity) during the last month preceding intervention start was 10.2 ± 3.0, 2.4 ± 2.3, and 0.9 ± 1.0 hours, respectively.

2.2 | Experimental design

The participants visited the test laboratory on three separate occasions. The first test session was a preliminary test used to determine VO_{2max} and maximal heart rate (HR_{max}) by a graded protocol to exhaustion and performing familiarization to cycling with vibration. On the subsequent two visits, time ≥90% of VO_{2max}, average power output during the work intervals, time ≥90% HR_{max}, electromyography (EMG) activity, blood lactate concentration ($[La^-]$), and rate of perceived exertion (RPE) was measured during the all-out HIT session with or without vibration during the 6 × 5 minutes work intervals with 2.5-minutes relief period in between. In order to say something about the recovery demand of these two HIT protocols, peak power output was measured during seated leg press before each session and 5 minutes after the last work interval. The order of interval protocols on the second and third test session was randomized and it was 48 ± 1 hours between these two test sessions.

2.3 | Preliminary test

A 20-minutes individual warm-up was performed before the start of the graded protocol to exhaustion for determination of VO_{2max}. This test has been described elsewhere.[15] Briefly, the test was initiated with 1 minutes of cycling at a power output corresponding to 3 W/kg (rounded down to the nearest 50 W). Power output was subsequently increased by 25 W every minute until exhaustion. Maximal aerobic power was calculated at the mean power output during the last minute of the test, and VO_{2max} was calculated as the average of the two highest 30 seconds VO_2 measurements by using a computerized metabolic system with mixing chamber (Oxycon Pro, Erich Jaeger, Hoechberg, Germany). The gas analyzers were calibrated with certified calibration gases of known concentrations before every test. The flow turbine (Triple V, Erich Jaeger, Hoechberg, Germany) was calibrated before every test with a 3 L calibration syringe (5530 series; Hans Rudolph, Kansas City, USA). The same metabolic system with identical calibration routines was used on the two subsequent experimental test sessions. HR was measured using a Polar RCX5 heart rate monitor (Polar, Kempele, Finland), and the highest value observed during the incremental exercise test was used as HR_{max}. Ten minutes after the end of the incremental exercise, subjects underwent a familiarization trial. They performed two 5-minutes work intervals with

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vibration at the intensity they expected to use during the experimental protocols.

2.4 | Experimental protocol

The cyclists were instructed to refrain from all types of intense exercise the day preceding a test session. They were also instructed to consume the same type of meal before each session and were not allowed to eat during the hour preceding a test or to consume coffee or other products containing caffeine during the 4 hours preceding the sessions. Both sessions were performed under similar environmental conditions (18-21°C) with a fan ensuring circulating air around the cyclist. Strong verbal encouragement was always given to ensure maximal effort. Both sessions for the individual cyclist were performed at the same time of day (±1 hour) to avoid influence of circadian rhythm. All testing was performed on the same bike, which was adjusted according to each cyclist's preference for seat height and horizontal distance between tip of seat and bottom bracket. Identical seating positions were used at both sessions. In order to monitor the power output during both HIT sessions, the bike was equipped with a PowerTap SL 2.4 power meter (CycleOps, Madison, USA) mounted on the rear wheel. The PowerTap device is a valid and reliable power meter during both vibration and no-vibration conditions.[16] In order to give resistance to the back wheel, the bike was connected to a Computrainer LabTM electromagnetically braked ergometer (Race Mate, Seattle, WA, USA), calibrated according to the manufacturer's specifications, which was attached to a vibration plate (Pneu-Vibe Pro; Pneumex, Inc. Sandpoint, ID, USA). The vibration frequency was set to 40 Hz (the 40 Hz frequency has been factory-validated to ±2 Hz). The vibration plate has been validated to generate an amplitude of 3 mm at an external load of 100, 200, and 300 kg.[17] The vibration platform produced the vibration stimulus in a synchronous way, meaning that the right and left side moved up and down at the same time. The test setup was inspired by Figure 1 in Munera et al[12] but instead of adding vibration to the front wheel with the dampening effect of the tire, we added vibration to the back frame that was connected to the Computrainer ergometer that was mounted to the vibration plate.

The two HIT protocols were always preceded by a test of lower-body power before the standardized warm-up to the HIT protocol started. The cyclists performed a 20-minutes individual warm-up (including cycling for 3 minutes at 200 W for normalization data for the EMG measurements) with a gradually increasing intensity. Thereafter, 1-minutes recovery was given before cycling 1 minutes at the planned power output of the first work interval followed by 2-minutes recovery before the first work interval started. The choice of work intervals lasting 5 minutes was based on the often reported improvement of performance in endurance-trained athletes after these intervals.[18-20] Both the TRAD and the VIB session were performed with the aim of highest possible average power output across all work intervals and the cyclists adjusted the power output themselves. The cyclists were familiar with 6 × 5-minutes exercise protocol, partly due to performing many HIT sessions with a similar design and partly due to their high training status and therefore rather good experience with optimizing their distribution of power output during such a task (as evident by a rather stable power output from the first to the last work interval). Power output during the relief periods was 50% of the power output used during work intervals. The cyclists accelerated to their preferred cadence and settled the starting power output using the Computrainer controller over the final 5 seconds of each rest period. During the final 5 seconds before start of each work interval, the vibration was applied in the VIB session. Oxygen uptake and HR were recorded at 15-second intervals throughout the work intervals. Time ≥90% VO_{2max} was calculated as the sum of VO_2 values (averaged over 15 seconds) that are superior or equal to 90% of the reference value for VO_{2max} obtained in the incremental exercise test to exhaustion. The same procedures were used for time ≥90% HR_{max}. Immediately after each work interval was venous blood sample from the fingertip analyzed for [La^-] (Biosen C-line, EKF Diagnostics, Barleben, Germany) and RPE was recorded using Borg's 6-20 scale.[21] Total VO_2 during each HIT session was calculated as the sum of the total oxygen uptake

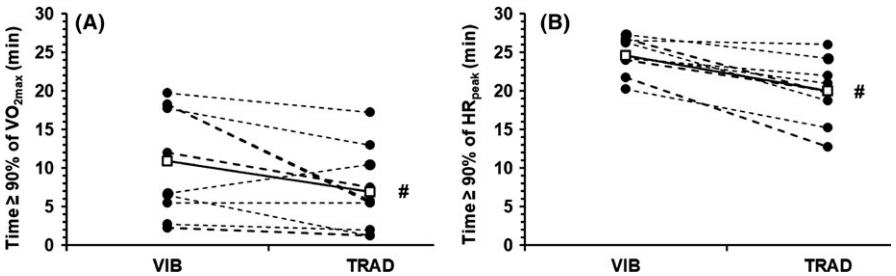

FIGURE 1 Individual data points (circle) and mean values (square) for time ≥90% of maximal oxygen uptake (VO_{2max}; panel A, and time ≥90% of peak heart rate (HR_{peak}; panel B, during a 6 × 5-min high-intensity aerobic interval training (HIT) session with vibration (VIB) or without vibration (TRAD). #Significant difference between treatments ($P < 0.05$)

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during the work intervals. To evaluate the development of VO_2 during the work intervals, the percentage of VO_{2max} accumulated during the first and last half of every work interval was calculated and the mean values across VIB and TRAD were compared.

2.5 | Peak power in leg press

Lower-body peak power was measured before and after both VIB and TRAD session using a pneumatic bilateral seated leg press machine (Keiser A420, Keiser Sports Health Equipment Inc, Fresno, CA, USA). The pneumatic equipment utilizes cylinders pressurized with air to provide variable resistance. This equipment has been described earlier and has been found to have a high reproducibility.[22] Briefly, software engineered for this equipment calculated work and power during the concentric phase of each repetition by sampling system pressure (force) and position at 400 Hz. The measurement before the HIT session was taken after 10 minutes of cycling at a RPE of 11-12 on the 6-20 Borg scale and was repeated 5 minutes after the last work interval of the HIT session. The cyclists sat with knees flexed at 90°-96° and the hips flexed at approximately 45° with the individual seating position being identical at all tests. After the 10 minutes of cycling, ten warm-up repetitions at submaximal and standardized load (100 kg) were performed before two repetitions where they extend their legs maximally and "as fast as possible." The power testing consisted of a single trial of ten external and standardized loads starting with 41 kg and ending with 249 kg. The recovery between loads was also standardized and increased from 5 seconds between the two lowest loads until 45 seconds between the two highest loads, ending up with a total duration of approximately 3 minutes from the first to last lift. The cyclists were instructed to exert force "as fast as possible" during all lifts. Average mechanical power output of both legs from the concentric phase of each lift was calculated and based on these were peak power output calculated by the manufacturer's software and used for statistical analysis. Only the leg press-specific warm-up was performed for the testing after the HIT sessions.

2.6 | Electromyography of m. vastus lateralis

The surface EMG bipolar electrodes (11 mm contact diameter and a 20-mm center-to-center distance) were positioned over the left m. vastus lateralis muscle belly parallel to muscle fibers orientation (2/3 of the distal distance between the spina iliaca anterior superior and the lateral side of the patella) according to recommendations to the SENIAM.[23] Before the placement of the gel-coated self-adhesive electrodes (Dri-Stick Silver Circular sEMG Electrodes AE-131, Neuro Dyne Medical, USA), the skin was shaved, abraded with emery paper, and cleaned with alcohol. The EMG module and the belonging signal cables were also taped to the thigh to reduce movement artefacts. A permanent marker was used to secure same electrode placement between the two test days.

Neuromuscular activation was measured with surface EMG electrodes (MuscleLab 4020e, Ergotest Technology AS, Langesund, Norway). The recorded EMG-signals were sampled at 1500 Hz, full waved rectified and filtered with a 5th order Butterworth low-pass filter using a cutoff frequency of 400 Hz and high- pass filter using a cutoff frequency of 20 Hz and smoothed over 100 ms with subsequent calculation of the moving average (MA). MA was averaged for 40 working cycles recorded between 2 minutes and 2.5 minutes from each of the six intervals. The average EMG signal for both conditions was normalized to 40 working cycles while cycling at 200 W without vibration. The EMG/power output ratio was calculated by dividing the average normalized MA value by the average power output for each of the six work intervals. The EMG data were processed in R[24] with the biosignalEMG package.[25]

2.7 | Statistics

All values presented in the text, figures, and tables are mean ± SD. Two-way repeated-measures ANOVA with Bonferroni post hoc tests were performed to evaluate differences between VIB and TRAD in the percentage of VO_{2max} accumulated during the first and last half of the work intervals (time vs conditioning). Student`s two-tailed paired *t* tests were used to evaluate potential differences between VIB and TRAD in physiological responses. The initial statistical significance level was set to $P \leq 0.05$; however, Holm-Bonferroni sequential adjustments were applied due to multiple comparisons.[26] ANOVA analyses were performed in GraphPad Prism 7 (GraphPad Software Inc., CA, USA) and Student's *t* tests were performed in Excel 2010 (Microsoft Corporation, Redmond, WA, USA). To assess differences in EMG activity between conditions, a linear mixed-effect model was fitted in the lme4 package.[27] Normalized EMG was the dependent variable in the model and intervals and conditions, as well as their interaction term, were set as fixed effects. Subjects were set with a random intercept. Inspection of residuals showed no deviance from the assumption of normal distribution. The EMG analysis was conducted in R.[24] Effect size of VIB was calculated with the following formula: ([VIB mean-TRAD mean]/TRAD SD). The scale proposed by Rhea[28] for highly trained subjects was used to interpret the magnitude of the treatment effect; 0.0-0.24 trivial, 0.25-0.49 small, 0.5-1.0 moderate, >1.0 large.

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3 | RESULTS

Responses during the two all-out HIT sessions are shown in Table 1. VIB induced a longer time ≥90% of VO_{2max} (ES = 0.77), longer time ≥90% of HR_{max} (ES = 1.13), and higher mean VO_2 (ES = 0.32) during the work intervals than TRAD (individual data from the two-first variables are presented in Figure 1, all $P < 0.05$). Mean percentage of VO_{2max} during the first 2.5 minutes across all work intervals was higher in VIB than TRAD (81.7 ± 2.8 vs 78.8 ± 2.6%, respectively, $P < 0.05$, ES = 1.12), while the values for last 2.5 minutes across all work intervals were not statistical significant different between VIB and TRAD (90.1 ± 2.9 vs 88.2 ± 2.6%, respectively; ES = 0.71, Figure 2). Averaged EMG activity in m. Vastus Lateralis across all work intervals was larger in VIB than TRAD ($P < 0.01$, ES = 0.43, Figure 3). There was no interaction effect between the individual intervals and the conditions ($P = 0.6$, Figure 3). The EMG/power output ratio across all work intervals was higher in VIB than TRAD ($P < 0.05$, ES = 0.36, Figure 3). Mean values across work intervals showed no difference between VIB and TRAD in mean power output, RPE, or [La⁻] (Table 1). There were no differences between VIB and TRAD in peak power in leg press either before (755 ± 141 vs 753 ± 140 W, respectively) or after (756 ± 146 vs 742 ± 164 W, respectively) the HIT session.

TABLE 1 Data from the 6 × 5-min high-intensity aerobic interval training (HIT) session with vibration during the work intervals (VIB) or without vibration (TRAD)

	VIB	TRAD
RPE_{peak}	19.6 ± 0.5	19.5 ± 0.5
RPE_{mean}	17.6 ± 0.8	17.4 ± 1.0
$[La^-]_{peak}$ (mmol/L)	13.1 ± 3.0	11.5 ± 2.8
$[La^-]_{mean}$ (mmol/L)	11.3 ± 3.3	11.2 ± 3.5
Mean power output (W)	352 ± 35	347 ± 43
VO_{2peak} (L/min)	5.33 ± 0.50	5.26 ± 0.42
VO_{2mean} (L/min)	5.00 ± 0.51*	4.86 ± 0.43
Time ≥90% of VO_{2max} (min)	10.99 ± 7.00*	6.95 ± 5.28
HR_{peak} (beats/min)	190 ± 5	191 ± 6
HR_{mean} (beats/min)	181 ± 4	178 ± 6
Time ≥90% of HR_{max} (min)	24.61 ± 2.38*	19.97 ± 4.12
EMG_{mean} (%-change from 200 W)	39.6 ± 19.3*	31.1 ± 19.7

Values are mean ± SD.
$[La^-]_{mean}$, mean blood lactate concentration after the work intervals; $[La^-]_{peak}$, peak blood lactate concentration; EMG_{mean}, mean electromyography activity across all six work intervals; HR_{mean}, mean heart rate during the work intervals; HR_{peak}, peak heart rate during the work intervals; RPE_{mean}, mean rate of perceived exertion after the work intervals; RPE_{peak}, peak rate of perceived exertion; VO_{2mean}, mean oxygen consumption during the work intervals; VO_{2peak}, peak oxygen consumption.
*Different from TRAD ($P < 0.05$).

4 | DISCUSSION

The main finding of the present study was that the VIB session induced a longer time above 90% of VO_{2max} and HR_{max} than the TRAD session. The superiority of VIB was accompanied by a larger averaged EMG activity in m. Vastus Lateralis across the work intervals and further supported by the effect size analyses showing a moderate-to-large practical effect of VIB on time above 90% of VO_{2max} and HR_{peak}. There were no differences between VIB and TRAD in mean power output, RPE or [La⁻] during the work intervals.

To the best of our knowledge, this is the first study to demonstrate longer time ≥90% of VO_{2max} when vibration is added to a HIT session in well-trained cyclists. This finding is supported by the two other studies investigating acute effects of adding vibration to cycling in the heavy exercise domain.[10,13] Although these two former studies were performed on recreational to moderate trained individuals

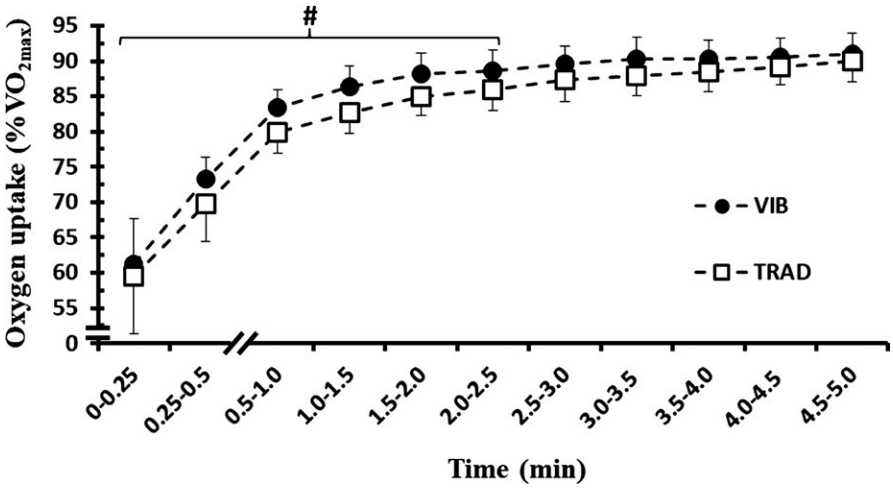

FIGURE 2 Development of mean VO_2 across all 6 work intervals during the high-intensity aerobic interval training (HIT) session with added vibration (VIB) and the session without vibration (TRAD). #Significant difference between treatments ($P < 0.05$)

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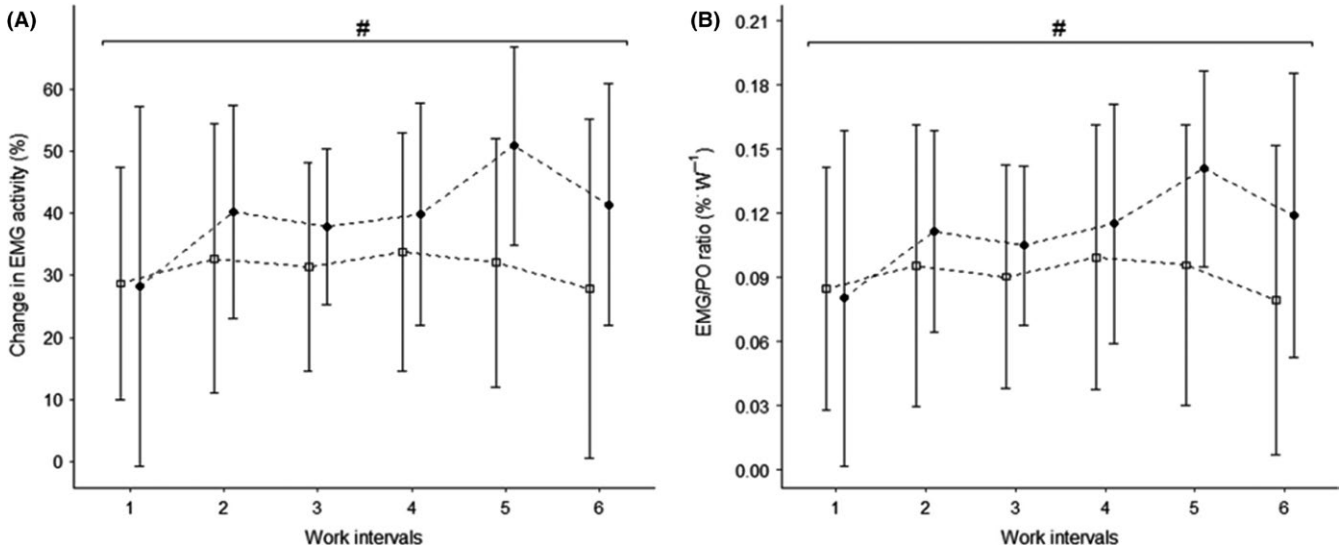

FIGURE 3 Panel A shows percentage change from 200 W in electromygraphic (EMG) activity of the m. Vastus Lateralis during a 6 × 5-min high-intensity aerobic interval training (HIT) session with vibration (black circles) or without vibration (white squares). Panel B shows the ratio between EMG and power output during the same session as panel A with vibration (black circles) or without vibration (white squares). #Significant main effect of conditions ($P < 0.05$)

and only one incremental test to exhaustion was performed, they observed higher VO_2 when vibration was added.[10,13] The longer time ≥90% of VO_{2max} in VIB may be connected to the observation of higher EMG activity in m. vastus lateralis and thus higher VO_2 demand. Indeed, it has been observed that a higher ratio between EMG and power output, indicating a higher neuromuscular cost, is related to longer time at high VO_2 during HIT intervals.[29] The latter is in agreement with the present finding where the EMG/power output ratio was higher in VIB than TRAD. Adding vibrations to the bike might increase muscle activity via increased activation of the stretch-sensitive muscle spindles ensuring increased Ia afferent signals and thus a larger excitatory inflow to the motoneurone pool and thereby increased muscle activation.[8,10,11] Interestingly, the only previous study investigating the effects of adding vibrations to low-intensity cycling observed a small, but significant increase in lower-body muscle activation.[12] The authors suggests that the low exercise intensity and the attenuated transmission of the vibration induced by adding vibrations to the front tire could explain no increase in VO_2 in their moderate trained cyclists.[12] Furthermore, it has been suggested that vibration-induced increase in muscle activity increases with higher exercise loads,[30] indicating a larger potential for increased muscle activation in the heavy exercise domain utilized during HIT.

Increased muscle activation could in theory increase the metabolic demand, thus increase VO_2. Consequently, both time ≥90% of HR_{peak} and mean VO_2 during all work intervals was higher in VIB than TRAD. The increase in VO_2 can also be explained by (a) the suggestion that vibration primarily increases recruitment of fast twitch muscle fibers,[31] which are thought to utilize more VO_2 than the slow twitch muscle fibers, (b) by the suggestion that previously exhausted muscle fibers may be additionally reactivated by vibration[32] and thereby increase VO_2, or (c) increased muscle activation in the upper body due the vibration.[33] It might therefore be suggested that adding vibration to a HIT session speed up the physiological responses of the cardiovascular system. A closer look at the VO_2 development during the two HIT sessions reveals that the mean percentage of VO_{2max} was similar during the first 15 seconds across all work intervals. However, when we look at the first 2.5 minutes, VO_2 was significantly higher in VIB than TRAD (82% vs 79%, respectively) and thus supports that vibrations can speed up the physiological responses. For the last 2.5 minutes across all work intervals, there was no statistically difference between VIB than TRAD (90% vs 88%, respectively), but the ES supported a superiority of VIB and showed a moderate effect of VIB (ES = 0.71). Moreover, it has been suggested that the training time ≥90% VO_{2max} can serve as a good criteria to judge the effectiveness of the stimulus to improve aerobic fitness in well-trained endurance athletes.[2,5,6] Since it has recently been observed that the HIT session that elicits the highest time at high VO_2 induces the largest training adaptation,[7] adding vibration to HIT cycling sessions could be a method to improve the quality and adaptations to HIT training.

Filingeri et al[13] also observed a speeding of the physiological responses when vibration was added to exercise. However, they used an incremental exercise test to exhaustion and notice an earlier development of fatigue with vibration, leading to exhaustion at a lower power output than without vibration. Furthermore, Samuelson et al[14] also observed that adding vibrations to cycling acutely reduced time to exhaustion from

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60 to 47 minutes in moderate trained individuals. To the contrary, the majority of data from the present study and the study of Sperlich et al[10] does not indicate earlier development of fatigue when adding vibration. This divergence might be related to the higher training status of the participants in the two latter studies. The primary measure of fatigue in the present study is the mean power output across all work intervals, but there was no difference between VIB and TRAD (352 vs 347 W, respectively). Another indicator of fatigue was the peak power in leg press performed 5 minutes after the last work interval, but no difference between VIB and TRAD was observed. Importantly, both the VIB and the TRAD session were performed with similar effort (as evident by similar RPE and [La^-] across all work intervals). This is further supported by no differences between VIB and TRAD in peak values of VO_2 and HR. However, it has been suggested that applying mechanical vibration may influence the activation of afferent input from sensory units in muscle fibers and attenuate pain sensation associated with exercise[34] and thus potentially mask a higher effort by VIB than TRAD. Interestingly, averaged EMG/power output ratio across all work intervals was larger in VIB than TRAD, indicating a decline of the neuromuscular efficiency during the VIB session. However, if the vibrations masked a higher effort, from a physiological perspective, one would expect higher blood lactate values, but this did not happen. Alternatively, adding vibration to cycling has been hypothesized to induce additional opening of blood vessels and thereby increase in blood distribution that could favor blood lactate elimination of lactate within the muscle tissue,[10] but this remains to be investigated.

No difference between VIB and TRAD in peak power during leg press 5 minutes after the last work interval also indicates that there should be no extra recovery demand with regard to contractile function in the lower body muscles after adding vibration to a HIT session. However, the present study did not investigate other important factors that need recovery, such as restoration of the glycogen stores, so a more comprehensive evaluation of long-term recovery demands of a longer time ≥90% of VO_{2max} due to VIB is needed. No negative effect of vibration on the development of indices of fatigue during exercise and postexercise recovery in the present study might be related the fact that highly trained cyclists were used, whereas participants in the other studies had a lower training status and were likely not cyclists (not clearly stated).

The vibration frequency is likely to affect the findings, since it has been observed that VO_2 during dynamic squatting increases in a linear fashion when the vibration frequency is increased from 18 to 34 Hz.[35] The present study used a vibration frequency of 40 Hz, while the former studies investigating the effects of adding vibration to cycling in the heavy exercise domain used a somewhat lower frequency of 20-33 Hz.[10,13] As with the other studies adding vibration to cycling,[10,12,13] the exact vibration characteristics induced to the cyclists in the present study are unknown. The vibrations have clearly an effect on the cyclists' acute physiological response to exercise, but the mechanical structure of the bike is likely to attenuate the vibration characteristics from the vibration platform to the lower limb. Future studies should look closer on the effects of different vibration characteristics.

In conclusion, the present study suggests that adding vibration to the work intervals during a HIT cycle session induced a significantly longer time >90% of VO_{2max} and HR_{peak} than the TRAD session. This may at least partly be due to the increased EMG activity in m. vastus lateralis. Furthermore, no negatively impact of VIB on the development of fatigue, mean power during the work intervals or recovery demands was observed in these well-trained cyclists.

5 | PERSPECTIVE

The importance of accumulated time ≥90% VO_{2max} to improve performance in well-trained endurance athletes is well established.[1,2,4,5] The results of the present study indicate that adding vibration to the work intervals during a cycling HIT session acutely increases time time ≥90% VO_{2max} and might therefore be recommended in order to optimize the exercise stimulus of well-trained cyclists. However, the long-term effects and training adaptations to this training method remain to be investigated. Furthermore, the long-term demands of recovery are also uncertain, so it is important to carefully monitor the training response and recover needs if this training method is used.

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